Indra R. Bishnoi, PhD

Email | Website | LinkedIn | Google Scholar

Current Position

2024– **Postdoctoral Research Fellow**
Lurie Center for Autism, Massachusetts General Hospital (MGH) and Harvard Medical School (HMS), Boston, Massachusetts, USA
Primary advisors: Dr. Evan A. Bordt, Dr. Nicole R. Zürcher
[1] Cell-type-specific outcomes of a high-risk gene for autism spectrum disorder: implications for brain and behavioral development.
[2] Juvenile sex differences in social interaction within a five-chamber sociability task following lipopolysaccharide exposure.
[3] Maternal autoimmunity's role in the pathophysiology underlying autism spectrum disorders and other neurodevelopmental disorders.

Education

2018 – 2023 Ph.D. Neuroscience (Behavioural Neuroscience)
Western University, London, Ontario, Canada
Primary advisors: Dr. Klaus-Peter Ossenkopp, Dr. Martin Kavaliers
[1] The effects of lipopolysaccharide exposure on the rodent model of lithium chloride-induced anticipatory nausea.
[2] Sex and age differences in the behavioural effects (spatial learning, startle response, social behaviour, anxiety-like behaviour and locomotor activity) of early adolescent two-hit lipopolysaccharide treatment in adolescent and adult rats.
[3] Neurobehavioural effects of di-2-ethylhexyl phthalate (DEHP) exposure in male and female rats: implications for obesity.

2013 – 2018 B.A. Honours Double Major, Psychology and Sociology
Western University, London, Ontario, Canada
Primary advisors: Dr. Klaus-Peter Ossenkopp, Dr. Martin Kavaliers
The effects of repeated administrations of propionic acid in juvenile rats: implications for autism spectrum disorders.

Research Funding (Role: PI)

2026– Pilot Funding Award: Massachusetts General Brigham ($60,000 USD)

2026– Post-Doctoral Save Our Scientists Grant Program: Autism Science Foundation ($25,000 USD)

2025– Confocal Microscopy Pilot Grant: Lurie Center for Autism ($1,950 USD)

Fellowships & Scholarships

2025 – 2028 Canada Postdoctoral Research Award: Canadian Institutes of Health Research (CIHR) ($210,000 CAD)
Rank: Top 7% of national applicants. Offer declined given previous acceptance of the Banting Postdoctoral Fellowship.

2025 – 2027 Sir Frederick G. Banting Postdoctoral Fellowship Program: CIHR ($140,000 CAD)
Rank: Top 10% of national applicants to the CIHR Selection Committee.

2024 – 2025 Landreth Research Fellowship: Robert and Donna Landreth Autism Research Fund ($70,000 USD)

2020 – 2023 Canada Graduate Scholarship-Doctoral Program: Natural Sciences and Engineering Research Council (NSERC) ($105,000 CAD)
Rank: Top 5% of applicants to the Psychology Review Committee.

2020 Ontario Graduate Scholarship ($15,000 CAD)
Declined given previous acceptance of the Canada Graduate Scholarship-Doctoral Program.

2019 Canada Graduate Scholarship-Master's Program: NSERC ($17,500 CAD)

2019 Ontario Graduate Scholarship ($15,000 CAD)
Declined given previous acceptance of the Canada Graduate Scholarship-Master's Program.

2015 Outstanding Student Scholarship: Secure-It ($1,000 CAD)

2013 Scholarship of Distinction: Western University ($1,500 CAD)

Awards & Honours

2025 Best Poster Presentation Travel Award: MassGeneral *for* Children Pediatric Research Day ($1,000 USD)
To attend the Psychoneuroimmunology Research Society 2025 annual meeting in Bordeaux, France.

2025 Top Scoring Fundamental Research Abstract: MassGeneral *for* Children Pediatric Research Day

2025 Travel Award: International Behavioral Neuroscience Society (IBNS) ($1,000 USD)

To attend the International Behavioral Neuroscience 2025 annual meeting in Tromsø, Norway.

2023 Top Cited Article Award from 2021-2022 in Developmental Psychobiology: Wiley

2023 Travel Award: Neuroscience Graduate Program ($1,000 CAD)
To attend the International Congress on Academic Medicine in Québec City, Canada.
Eligibility: Top 5% of Graduate Neuroscience Program.

2022 Top Cited Article Award from 2020-2021 in Developmental Psychobiology: Wiley

2019 Travel Award: Society of Neuroscience Graduate Students ($500 CAD)
To attend the Society for Neuroscience conference in Chicago, USA.

2018 Dean's Honour List: Western University

2017 Dean's Honour List: Western University

2017 Social Science Alumni Award: Western University ($1,500 CAD)
Eligibility: Top cumulative average of the year in the undergraduate psychology program.

2013 Ontario Scholar: Ministry of Education

Leadership & Outreach Recognition

2022 Clothing4Climate Changemaker: Let's Talk Science

2022 A Conversation with a Scientist: Society of Neuroscience Graduate Students Profiles in Research

2021 Youth Green Leader Award: London Environmental Network

2021 Leadership and Community Engagement Award: Society of Neuroscience Graduate Students ($200 CAD)

2021 Best Business Plan Winner: Graduate Student Innovations Scholars ($1,000 CAD)

2021 Spotlight of the Month: Society of Neuroscience Graduate Students

2021 Spotlight on Women & Girls in Science: Science Odyssey, Science Rendezvous

2020 Science Rendezvous Innovation Showcase "Jack of All Trades: Indra Bishnoi"

Publications

[21] Kavaliers, M., **Bishnoi, I. R.,** Matic, V., Ossenkopp, K.-P., & Choleris, E. (2026). Avoidance of Bacterial and Viral Threat is Affected by Infection Status in a Terrestrial Snail. *Behavioural Processes, 241,* 105414. https://doi.org/10.1016/j.beproc.2026.105414

Open Access Dataset: https://doi.org/10.20383/103.0936
Pre-print: http://dx.doi.org/10.2139/ssrn.6187940

[20] **Bishnoi, I. R.,** Matic, V., Wang, L., Ravichandran, C., Lake, G., Kavaliers, M., & Ossenkopp, K.-P. (2026). Timing-dependent immune modulation of learning in a long-interval rodent model of anticipatory nausea. *Behavioural Brain Research, 502,* 116026. https://doi.org/10.1016/j.bbr.2026.116026

[19] Matic, V., Huynh, M., **Bishnoi, I. R.,** Kavaliers, M., & Ossenkopp, K.-P. (2025). Sex differences in sucrose-induced locomotor sensitization and cross-sensitization with the D2/D3 agonist quinpirole in rats. *Behavioural Brain Research, 495,* 115771. https://doi.org/10.1016/j.bbr.2025.115771

Pre-print: https://doi.org/10.21203/rs.3.rs-5004624/v1

[18] *Slamin, M., ***Bishnoi, I. R.,** Bankowski, I., Norris, H. A., & Bordt, E. A. (2025). Where's my mom? Resilient maternal preference in post-weaning male and female mice within a multi-chamber social behavior task. *Brain, Behavior, and Immunity – Health, 47,* 101034. https://doi.org/10.1016/j.bbih.2025.101034 **co-first authorship.*

[17] **Bishnoi, I. R.,** & Bordt, E. A. (2025). Sex and Region-Specific Differences in Microglial Morphology and Function Across Development. *Neuroglia, 6*(1), 2. https://doi.org/10.3390/neuroglia6010002

[16] **Bishnoi, I. R.,** & Bordt, E. A. (2024). The Missing Half: Paternal Immune Activation's Role in Transgenerational Sex Differences. *Brain, Behavior, and Immunity, 120,* 288-289. https://doi.org/10.1016/j.bbi.2024.06.010

[15] **Bishnoi, I. R.,** Kavaliers, M., & Ossenkopp, K.-P. (2024). Lipopolysaccharide (LPS) attenuates the primary conditioning of lithium chloride (LiCl)-induced context aversion but not the secondary conditioning of context aversion or taste avoidance. *Behavioural Brain Research, 459,* 114800. https://doi.org/10.1016/j.bbr.2023.114800

[14] Kavaliers, M., Wah, D., **Bishnoi, I. R.,** Ossenkopp, K.-P., & Choleris, E. (2023). Disgusted snails, oxytocin, and the avoidance of infection threat. *Hormones and Behavior, 155,* 105424. https://doi.org/10.1016/j.yhbeh.2023.105424

Open Access Dataset: https://doi.org/10.20383/103.0787

[13] **Bishnoi, I. R.,** Kavaliers, M., & Ossenkopp, K.-P. (2023). Immune activation attenuates memory acquisition and consolidation of conditioned disgust (anticipatory nausea) in rats. *Behavioural Brain Research, 439,* 114250. https://doi.org/10.1016/j.bbr.2022.114250

[12] **Bishnoi, I. R.,** Cloutier, C. J., Tyson, C. D., Matic, V., Kavaliers, M., & Ossenkopp, K.-P. (2022). Infection, learning, and memory: Focus on immune activation and aversive conditioning. *Neuroscience and Biobehavioral Reviews, 142,* 104898. https://doi.org/10.1016/j.neubiorev.2022.104898

[11] Klebl, C., Donner, M., & **Bishnoi, I. R.** (2022). Disliking: From adaptive disgust to ugliness. In M. Skov & M. Nadal (Eds.), *The Routledge International Handbook of Neuroaesthetics* (1st ed.). Routledge. https://doi.org/10.4324/9781003008675

[10] Kavaliers, M., Ossenkopp, K.-P., Tyson C. D., **Bishnoi, I. R.,** & Choleris, E. (2022). Social factors and the neurobiology of pathogen avoidance. *Biology Letters, 18,* 20210371. http://doi.org/10.1098/rsbl.2021.0371

[9] Kavaliers, M., **Bishnoi, I. R.,** Ossenkopp, K.-P., & Choleris, E. (2022). Odor based mate choice copying in deer mice is not affected by familiarity or kinship. *Animal Cognition, 25,* 241–248. https://doi.org/10.1007/s10071-021-01550-z

[8] Kavaliers, M., **Bishnoi, I. R.,** Ossenkopp, K.-P., & Choleris, E. (2021). Progesterone and disgust: A response to "progesterone does raise disgust". *Hormones and Behavior, 137,* 104936. https://doi.org/10.1016/j.yhbeh.2021.104936

[7] Kavaliers, M., **Bishnoi, I. R.,** Ossenkopp, K.-P., & Choleris, E. (2021). Differential effects of progesterone on social recognition and the avoidance of pathogen threat by female mice. *Hormones and Behavior, 127,* 104873. https://doi.org/10.1016/j.yhbeh.2020.104873

[6] **Bishnoi, I. R.,** Ossenkopp, K.-P., & Kavaliers, M. (2021). Sex and age differences in locomotor and anxiety-like behaviours in rats: From adolescence to adulthood. *Developmental Psychobiology, 63,* 496–511. https://doi.org/10.1002/dev.22037

[5] Kavaliers, M., Colwell D. D., Wah, D., **Bishnoi, I. R.,** Ossenkopp, K.-P., & Choleris, E. (2019). Conspecific infection threat rapidly biases the social responses of female mice: Involvement of oxytocin. *Hormones and Behavior, 113,* 67–75. https://doi.org/10.1016/j.yhbeh.2019.04.016

[4] Wah, D., Ossenkopp, K.-P., **Bishnoi, I. R.,** & Kavaliers, M. (2019). Predator odor exposure in early adolescence influences the effects of the bacterial product, propionic acid, on anxiety, sensorimotor gating, and acoustic startle response in male rats in later adolescence and adulthood. *Physiology & Behavior*, *199,* 35–46. https://doi.org/10.1016/j.physbeh.2018.11.003

[3] Wah, D., Kavaliers, M., **Bishnoi, I. R.,** & Ossenkopp, K.-P. (2019). Lipopolysaccharide (LPS) induced sickness in early adolescence alters the behavioral effects of the short-chain fatty acid, propionic acid, in late adolescence and adulthood: Examining anxiety and startle reactivity. *Behavioural Brain Research*, *360,* 312-322. https://doi.org/10.1016/j.bbr.2018.12.003

[2] Kamen, C. L., Zevy, D. L., Ward, J. M., **Bishnoi, I. R.,** Kavaliers, M., & Ossenkopp, K.-P. (2019). Systemic treatment with the enteric bacterial fermentation product, propionic acid, reduces acoustic startle response magnitude in rats in a dose-dependent fashion: contribution to a rodent model of ASD. *Neurotoxicity Research, 35,* 353-359. https://doi.org/10.1007/s12640-018-9960-9

[1] **Bishnoi, I. R.** (2017). A call for the combined pharmacological and psychological treatment of depression. *Western Undergraduate Psychology Journal, 5*(1). http://ir.lib.uwo.ca/wupj/vol5/iss1/1

Doctoral Dissertation

[1] **Bishnoi, I. R.** (2023). *The effects of lipopolysaccharide (LPS) on conditioned disgust: Expanding the rodent model of lithium chloride (LiCl)-induced anticipatory nausea* [Doctoral dissertation, Western University]. *Electronic Thesis and Dissertation Repository*. https://hdl.handle.net/20.500.14721/37294

Other Scientific Writings

[2] **Bishnoi, I. R.** (2020). The Stressful Storm of Adolescence: Implications for Autism Spectrum Disorder. *The Dorsal Column*. https://songsuwo.ca/thedorsalcolumn/vol1-iss4-indra-bishnoi

[1] **Bishnoi, I. R.** (2020). The Nauseating Side of Chemotherapy. *The Dorsal Column*. https://songsuwo.ca/thedorsalcolumn/vol1-iss2-indra-bishnoi

Submitted Publications

[2] *Bankowski, I., ***Bishnoi, I. R.,*** Shin, E., Cashin, M., Norris H. A., Barry, D., Slamin, M., & Bordt, E. A. (2026). *Cx3cr1*-BAC-CRE-mediated knockout of toll-like receptor 4 alters mouse communicative behaviors, microglial morphology, and engulfment of synaptic material. *Manuscript submitted to Glia. *co-first authorship.*

[1] Huynh, M., Matic, V., **Bishnoi, I. R.,** Lake, G., Ammar Al-Khudairy, M., Kavaliers, M., & Ossenkopp, K.-P. (2026). *The effect of estrogen and progesterone replacement on conditioned disgust (anticipatory nausea) in ovariectomized female rats. Manuscript submitted to Pharmacology Biochemistry and Behaviour.*

Publications In Preparation

[1] **Bishnoi, I. R.,** Ossenkopp, K.-P., & Kavaliers, M. (2026). *Open-field behaviours in adolescent rats: Influence of sex and immune activation on variable reliabilities and intercorrelations. Manuscript in preparation for Physiology & Behavior.*

Teaching & Mentoring

Directly supervised 21 trainees (9 undergraduates, 5 technicians, 7 MSc/PhD students), 2018 – Present. Trainees have co-authored 8 publications, 18 conference presentations, and received 5 scholarships.

2026	Lurie Center Summer Internship Program (Mentor) Mentored a summer intern through the analyses of several rodent behavioral assessments. At the conclusion of this program, the intern successfully defended a poster on the research topic.	Lurie Center for Autism
2026	Mucosal Immunology and Biology Research Center (Mentor) Mentored two students on neurodevelopmental disorders (NDDs) and gut cell gene regulation. At the conclusion of this program, the students successfully completed an oral presentation on the topic.	Massachusetts General Hospital
2025	Teaching Institute (Selected Participant) Four-day intensive program in which I refined lesson plans for future neuropsychology courses, developed and delivered new content to fellow postdoctoral scholars, and exchanged structured peer feedback to refine teaching practices.	Harvard Medical School
2024	Cognitive Science (Psych. 3139B, Adjunct Assistant Professor)	Western University
2024	Neuropsychology and Cognitive Neuroscience (Psych. 3224B, Adjunct Assistant Professor)	Western University
2023	Neuropsychology and Cognitive Neuroscience (Psych. 3224B, Lecturer)	Western University
2021 – 2022	Introduction to Psychology (Psych. 1015B, Teaching Assistant (TA))	Brescia College
2020 – 2022	Introduction to Psychology (Psych. 1010A, TA)	Brescia College
2020	Introduction to Animal Cognition (Psych. 2210B, TA)	Western University
2019	Sex Differences in Human Brain and Behaviour (Psych. 3225B, TA)	Western University
2019	“Studying Sex Differences Using Current Research Methods in Behavioural Neuroscience” (Invited Guest Lecturer)	Western University
2018	Evolution and Psychology: The Science of Human Nature (Psych. 3228A, TA)	Western University
2018	Bacterial Metabolites and Brain and Behaviour (Guest Lecturer)	Kiwanis Senior’s Centre

Peer & Grant Reviews

2026	*Neurotoxicity Research,* *Behavioral and Brain Functions*
2025–	*CIHR Doctoral Research Awards Committee Member*
2025	*Animal Behaviour,* *Physiology & Behavior*
2024	*Brain, Behavior, and Immunity* (assisted), *Neuropsychopharmacology* (assisted), *Frontiers in Behavioural Neuroscience* (Behavioral Endocrinology, Motivation and Reward sections), *Frontiers in Psychology* (Cognitive Science section), *Behavioural Pharmacology*
2019 – 2020	*University of Western Ontario Medical Journal*

Conferences and Presentations (*presenter)

International

[18] ***Bishnoi, I. R.,*** Slamin, M. M., Bankowski, I. M., Shin, E. E., Barry, D. T., Mandal, S., Norris, H. A., Bordt, E. A. (2026, Jun). *Characterizing cell-type-specific contributions of a high-penetrance gene for autism spectrum disorder: Implications for brain, gut, and behavioral development* [Invited oral presentation]. Gordon Research Conference, Maine, USA.

[17] ***Bishnoi, I. R.,*** Slamin, M. M., Bankowski, I. M., Shin, E. E., Barry, D. T., Mandal, S., Norris, H. A., Bordt, E. A. (2026, Jun). *Characterizing cell-type-specific contributions of a high-penetrance gene for autism spectrum disorder: Implications for brain, gut, and behavioral development* [Poster presentation]. Gordon Research Seminar/Conference, Maine, USA.

[16] ***Bishnoi, I. R.,*** Bankowski, I. M., Slamin, M. M., Norris, H. A., Bordt, E. A. (2025, Jun). *Maternal TLR7 activation increases maternal autoimmunity resulting in male-biased ASD-like behaviors* [Invited oral presentation]. International Behavioral Neuroscience Society, Tromsø, Norway.

[15] ***Bishnoi, I. R.,*** Bankowski, I. M., Slamin, M. M., Norris, H. A., Bordt, E. A. (2025, Jun). *Maternal TLR7 activation increases maternal autoimmunity resulting in male-biased ASD-like behaviors* [Poster presentation]. International Behavioral Neuroscience Society, Tromsø, Norway.

[14] ***Bishnoi, I. R.,*** Bankowski, I. M., Slamin, M. M., Norris, H. A., Bordt, E. A. (2025, Jun). *Maternal TLR7 activation increases maternal autoimmunity resulting in male-biased ASD-like behaviors* [Poster presentation]. Psychoneuroimmunology Research Society, Bordeaux, France.

[13] ***Bishnoi, I. R.,*** Kavaliers, M., Ossenkopp, K.-P. (2023, Oct). *The effects of lipopolysaccharide (LPS) on conditioned disgust: Expanding the rodent model of lithium chloride (LiCl)-induced anticipatory nausea* [Invited oral presentation]. Institute of Medical Psychology, Essen, Germany, Satellite.

[12] ***Bishnoi, I. R.,*** Matic, V., Kavaliers, M., Ossenkopp, K.-P. (2023, Jun). *Attenuation of lithium chloride-induced anticipatory nausea (AN) following immune activation* [Poster presentation]. International Behavioral Neuroscience Society, Niagara Falls, Canada.

[11] *Wah, D., ***Bishnoi, I. R.,*** Kavaliers, M., Ossenkopp, K.-P., Choleris, E. (2023, Jun). *Disgust-like responses to infection threat and their modulation in a land snail* [Poster presentation]. International Behavioral Neuroscience Society, Niagara Falls, Canada.

[10] *Matic, V., ***Bishnoi, I. R.,*** Kavaliers, M., Ossenkopp, K.-P. (2023, Jun). *The effect of sucrose pre-exposure on quinpirole-induced locomotor sensitization: An investigation of sex differences in adult rats* [Poster presentation]. International Behavioral Neuroscience Society, Niagara Falls, Canada.

[9] ***Bishnoi, I. R.,*** Kavaliers, M., Ossenkopp, K.-P. (2023, Apr). *Immune activation impairs the acquisition and consolidation of lithium chloride-induced anticipatory nausea in a rodent model* [Poster presentation]. International Congress on Academic Medicine, an initiative by the Association of Faculties of Medicine of Canada, Québec City, Canada.

[8] ***Bishnoi, I. R.,*** Ossenkopp, K.-P., Kavaliers, M. (2022, Nov). *Lipopolysaccharide-induced impairment of memory acquisition and consolidation processes in a rodent model of lithium chloride-induced anticipatory nausea* [Poster presentation]. Society for Neuroscience, San Diego, USA.

[7] *Matic, V., ***Bishnoi, I. R.,*** Kavaliers, M., Ossenkopp, K.-P. (2022, Nov). *Estrogen replacement impairs the learning of environmentally conditioned disgust behaviour (anticipatory nausea) in ovariectomized female rats* [Poster presentation]. Society for Neuroscience, San Diego, USA.

[6] *Kavaliers, M., ***Bishnoi, I. R.*** (2021, Feb). *Group bias in non-human animals* [Invited oral presentation]. Society for Social Neuroscience, Satellite.

[5] ***Bishnoi, I. R.,*** Ossenkopp, K.-P., Kavaliers, M. (2021, Jan). *Open-field behaviors in adolescent rats: Influence of sex and immune activation on variable reliabilities and intercorrelations* [Poster presentation]. Society for Neuroscience, Satellite.

[4] ***Bishnoi, I. R.,*** Ossenkopp, K.-P., Kavaliers, M. (2020, Aug). *Rodent sex and age differences in locomotor and anxiety-like behaviors: From adolescence to adulthood* [Poster presentation]. International Behavioral Neuroscience Society, Satellite.

[3] ***Bishnoi, I. R.,*** Kavaliers, M., Ossenkopp, K.-P. (2019, Oct). *The behavioural effects of early adolescent lipopolysaccharide administration on adolescent and adult male and female rats* [Poster presentation]. Society for Neuroscience, Chicago, USA.

[2] ***Bishnoi, I. R.,*** Kavaliers, M., Ossenkopp, K.-P. (2019, Oct). *The behavioural effects of early adolescent lipopolysaccharide administration on adolescent and adult male and female rats* [Poster presentation]. Society for Social Neuroscience, Chicago, USA.

[1] *Kavaliers, M., ***Bishnoi, I. R.,*** Ossenkopp, K.-P., Choleris, E. (2019, Feb). *Progesterone and pathogen elicited disgust* [Poster presentation]. Steroids and Nervous System, Turin, Italy.

National and Regional

[19] ***Bishnoi, I. R.,** Slamin, M. M., Bankowski, I. M., Shin, E. E., Barry, D. T., Mandal, S., Norris, H. A., Bordt, E. A. (2026, May). *Characterizing cell-type-specific contributions of a high-penetrance gene for autism spectrum disorder: Implications for brain, gut, and behavioral development* [Poster presentation]. Canadian Association for Neuroscience, Montréal, Canada.

[18] *Shin, E. E., *Barry, D. T., Bankowski, I. M., Slamin, M. M., Norris, H. A., **Bishnoi, I. R.,** Bordt, E. A. (2026, May). Early life adversity alters offspring behavior via microglial toll-like receptor 4 signaling [Poster presentation]. MassGeneral for Children Pediatric Research Day, Boston, USA.

[17] ***Bishnoi, I. R.,** Slamin, M. M., Bankowski, I. M., Shin, E. E., Barry, D. T., Mandal, S., Norris, H. A., Bordt, E. A. (2026, Apr). *Characterizing cell-type-specific contributions of a high-penetrance gene for autism spectrum disorder: Implications for brain, gut, and behavioral development* [Invited poster presentation]. Lurie Center for Autism Scientific Retreat, Boston, USA.

[16] *Huynh, M., Matic, V., **Bishnoi, I. R.,** Ossenkopp, K.-P., Kavaliers, M. (2025, May). *Combined estrogen and progesterone treatment enhances the learning of anticipatory nausea in ovariectomized female rats* [Poster presentation]. Southern Ontario Neuroscience Association, London, Canada.

[15] ***Bishnoi, I. R.,** Slamin, M. M., Bankowski, I. M., Norris, H. A., Bordt, E. A. (2025, Apr). *Characterizing cell-type-specific contributions of a high-penetrance gene for autism spectrum disorder: Implications for brain and behavioral development* [Invited poster presentation]. Lurie Center for Autism Scientific Retreat, Boston, USA.

[14] ***Bishnoi, I. R.** (2025, Apr). *Characterizing cell-type-specific contributions of a high-penetrance gene for autism spectrum disorder: Implications for brain and behavioral development* [Invited oral presentation]. MassGeneral for Children Pediatric Research Day, Boston, USA.

[13] ***Bishnoi, I. R.,** Slamin, M. M., Bankowski, I. M., Norris, H. A., Bordt, E. A. (2025, Apr). *Characterizing cell-type-specific contributions of a high-penetrance gene for autism spectrum disorder: Implications for brain and behavioral development* [Poster presentation]. MassGeneral *for* Children Pediatric Research Day, Boston, USA.

[12] *Bankowski, I. M., Slamin, M. M., **Bishnoi, I. R.,** Norris, H. A., Bordt, E. A. (2025, Apr). Maternal TLR7 activation induces maternal autoimmunity, resulting in male-biased immune and ASD-like behavior alterations [Poster presentation]. MassGeneral for Children Pediatric Research Day, Boston, USA.

[11] *Slamin, M. M., *McDaniel, M., **Bishnoi, I. R.,** Bankowski, I. M., Norris, H. A., Bordt, E. A. (2025, Apr). The unbreakable relationship between mothers and juveniles: Maternal preference in a five-chamber social behavior task [Poster presentation]. MassGeneral for Children Pediatric Research Day, Boston, USA.

[10] *Bankowski, I. M., Slamin, M. M., **Bishnoi, I. R.,** Norris, H. A., Bordt, E. A. (2025, Apr). Maternal TLR7 activation induces maternal autoimmunity, resulting in male-biased immune and ASD-like behavior alterations [Poster presentation]. Massachusetts General Hospital Celebration for Science, Boston, USA.

[9] *Slamin, M. M., **Bishnoi, I. R.,** Bankowski, I. M., Norris, H. A., Bordt, E. A. (2025, Apr). The unbreakable relationship between mothers and juveniles: Maternal preference in a five-chamber social behavior task [Poster presentation]. Massachusetts General Hospital Celebration for Science, Boston, USA.

[8] ***Bishnoi, I. R.** (2024 & 2025, Jul). *Lurie Center Interns Lunch and Learn* [Invited oral presentation]. Lurie Center for Autism, Lexington, USA.

[7] ***Bishnoi, I. R.** (2019, Sep). *Psychology graduate student association* [Invited oral presentation]. Fall Departmental Orientation, London, Canada.

[6] ***Bishnoi, I. R.,** Kavaliers, M., Ossenkopp, K.-P. (2019, May). *The behavioural effects of lipopolysaccharide in adolescent male and female rats* [Poster presentation]. Canadian Association for Neuroscience, Toronto, Canada.

[5] ***Bishnoi, I. R.,** Kavaliers, M., Ossenkopp, K.-P. (2019, Apr). *The behavioural effects of early adolescent lipopolysaccharide exposure on adolescent and adult male rats* [Poster presentation]. Southern Ontario Neuroscience Association, Windsor, Canada.

[4] ***Bishnoi, I. R.,** Kavaliers, M., Ossenkopp, K.-P. (2019, Apr). *Lipopolysaccharide induced behavioural effects in adolescent and adult male rats* [Poster presentation]. London Health Research Day, London, Canada.

[3] ***Bishnoi, I. R.** (2018, Sep). *Bacterial metabolites and brain and behaviour* [Oral presentation]. Kiwanis Senior's Centre, London, Canada.

[2] ***Bishnoi, I. R.,** Kavaliers, M., Ossenkopp, K.-P. (2018, May). *The effect of repeated administrations of propionic acid in juvenile rats: Implications for Autism Spectrum Disorder* [Poster presentation]. Southern Ontario Neuroscience Association, Hamilton, Canada.

[1] ***Bishnoi, I. R.** (2018, May). *The effect of repeated administrations of propionic acid in juvenile rats: Implications for Autism Spectrum Disorder* [Oral presentation]. Developmental Disabilities Research Day, London, Canada.

Other Presentations and Panels

2024	Rooted Resilience: Eco-Distress & Messy Hope (Invited Panelist)	Western University
2022	Curious Careers: Climate Changemakers (Invited Panelist)	Let's Talk Science
2022	Steps Towards a Sustainable Lifestyle (Invited Panelist)	WaterAid
2022	Sustainable Shopping Workshop: Informed by Research (Invited Speaker)	Huron College

Other Research & Clinical Experience

2024–	Clinical Shadow Observing consultations, neuropsychological assessments, and feedback sessions by psychiatrists and psychologists with children with autism and their families. Aiming to improve the translation and reverse translation between my preclinical research program and clinical diagnostic and treatment practices.	Lurie Center for Autism
2017	Dr. Klaus-Peter Ossenkopp & Dr. Martin Kavaliers Lab Conducted a behavioural research project using the acoustic startle and light-dark tasks. Analyzed data using SPSS. Created figures using Inkscape. Published findings in 2019.	Western University
2016 – 2020	Psychiatric Research Assistant, Dr. Carol McDaniel Studied the influence of vitamin D deficiency on forensic psychiatric inpatient health outcomes.	St. Joseph's Health Care London
2016	Cohen Lab, Dr. Adam Cohen Recruited, collected, and analyzed data from 3-6-year-old male and female children on a theory of mind task. Built strong relationships with children and families for participant-experimenter rapport.	Western University
2015 – 2016	CoNCH Lab, Dr. Ingrid Johnsrude Collected data using the electroencephalogram (EEG) and trained research assistants on EEG protocols. Trained on audiometry and eye-tracking, in addition to Qualtrics and MATLAB.	Western University

Select Work Experience

2021 – 2023	Founder Managed a team of 5. Communicated current climate science to a layperson audience, closing the science-action gap. Social media marketing resulting in a 70%+ monthly boost in reach.	Jeo by Indra
2019 – 2021	Sustainability Committee Chair Created an actionable climate emergency declaration at Western University in solidarity with the City of London. Campus-wide and London community outreach for environmental sustainability.	Western University
2018 – 2023	Lab Manager Development and scheduling of research projects, procurement of supplies, and maintenance of all ethical, security protocols, and standards. Additionally, providing mentorship to all students (20+), including project management, data analysis, academics, and financial aid procurement.	Western University

Select Community and Volunteer Experience

2026–	Travel Award Pre-Review Program: Mentor Assisting junior scientists (undergraduates, graduate students, and postdocs) in crafting competitive Travel Award applications to the International Behavioral Neuroscience Society Annual Meeting by providing feedback on their applications.
2025–	Menstrual Products Access Program: Organizer Established and maintaining a free menstrual products station by coordinating purchasing, restocking, and engaging the community to sustain access.
2025–	Membership Committee: Member & Co-Chair Contributing to the growth of the International Behavioral Neuroscience Society community by identifying, enhancing, and communicating the value of society membership to past, current, and prospective members.
2024–	Science Communication Subcommittee: Member Introducing postdocs at MGH and HMS to the resources needed to effectively communicate their science including public speaking (scientific and lay presentations/elevator pitches) and scientific writing opportunities (blogs/manuscripts/grants).
2024 – 2025	Community Engagement Subcommittee: Co-Chair Bringing together postdocs by organizing events focused on building community, sharing resources, and navigating challenges like financial aid procurement. Our goal is to create opportunities for connection, support, and professional growth throughout the postdoctoral experience at MGH.

2023	Animal-Based Research Transparency Initiative Coalition: Graduate Student Representative Mandated coalition to increase transparency regarding the ethical, purposeful, and regulated performance of animal-based research and teaching at Western University. Ensured that the relevant materials, training, and other supports were available to support buy-in from Western's research community.
2022 – 2024	University Consulting Group: Client Relations Associate, Team Lead, Consultant Led teams to conduct internal and external research to provide actionable deliverables for our clients. Aided our clients in understanding how to enter, develop, and thrive in the Canadian market, including post-COVID-19 employee satisfaction.
2020 – 2024	President's Advisory Committee on Environment & Sustainability: Designate Represented the Society of Graduate Students to advise Western University's President, Vice-Presidents, and Provost on how to best design and execute campus-wide climate & sustainability strategies and improve awareness of sustainability initiatives.
2019 – 2021	The Dorsal Column: Science Writer, Community Liaison Writer for quarterly neuroscience-related blog posts. Further, responsible for communications between the Dorsal Column team and the greater London community.
2019 – 2020	Society of Neuroscience Graduate Students: Mentor Upper-year academic and social mentor for incoming students to the neuroscience graduate program.
2018 – 2022	Society of Graduate Students: Neuroscience Councillor Advocacy and communication between Society of Graduate Students and the neuroscience program.
2018 – 2020	Canadian Association for Girls in Science: London Chapter Organizer Created and hosted events for girls aged 7-16 to promote, educate, and build confidence regarding current topics in science, technology, engineering, and mathematics.

Other Skills, Workshops, and Interests

[9] Statistics and Scientific Computing with R, Python, SPSS, MATLAB, Qualtrics, and GraphPad Prism

[8] SafeTALK: Suicide Alertness Training

[7] Teaching Assistant Training Program

[6] Animal Care & Veterinary Services Certification

[5] WestNIRS - fNIRS Workshop 2022, 2023

[4] Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans

[3] Grant Writing and Research Staff Management Training

[2] Woodworking, Swimming, Yoga, Meditation

[1] Language learning (Hindi, Punjabi, Urdu, French)